

Kahan Shah

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EDUCATION

University of Southern California

Master of Science in Computer Engineering

Los Angeles, CA

August. 2025 – Present

University of California, Santa Cruz

Bachelor of Science in Computer Engineering

Santa Cruz, CA

September. 2021 – June. 2025

EXPERIENCE

The Difference

Full Stack Intern

May 2024 – June 2025

Remote

- Architected and executed migration from legacy system to Laravel API, reducing response times by 40% and improving reliability
- Developed RESTful Laravel endpoints with Nutritionix integration, enabling real-time nutritional processing for 1000+ daily users
- Optimized Flutter frontend with debounced search, reducing server load by 60% and improving UX responsiveness
- Implemented Riverpod state management, eliminating redundant API calls and reducing backend traffic by 50%

University of California

Grader/Reader for Software Development Course

Sep. 2024 – June 2025

Santa Cruz, CA

- Evaluated programming assignments and delivered actionable feedback to 100+ students, reinforcing software development best practices

PROJECTS

Embedded/IoT Projects | *ESP32, C, FreeRTOS, BLE, HTTP*

- Built Bluetooth HID mouse controller on ESP32-C3 (ESP-IDF, BLE); integrated ICM-42670-P IMU and tilt-to-cursor mapping; achieved <10ms latency
- Developed ESP32 weather monitor (C, HTTP): parsed wttr.in API + local sensors and posted readings at 1 Hz
- Designed ultrasonic range finder (HC-SR04) with temperature compensation, improving accuracy by 95% in 10–20 cm range
- Implemented Morse-code optical link: Raspberry Pi LED transmitter + ESP32 ADC receiver; optimized sampling to increase throughput by 300%

ASIC/FPGA and Hardware Projects | *Verilog, SystemVerilog, FPGA, RISC-V, Cadence Virtuoso*

- Built RISC-V assembly graphics routines and an interactive “Paint” app (lines/circles/polygons, input handling), improving frame time by 40%
- Created gate-level schematics and physical layouts in Cadence Virtuoso for a 16-bit MAC and key blocks (Booth encoder, 6-2 compressor, CSA/CLA adders, DFFs); ran DRC/LVS and iterative optimization for power, delay, and area.
- Designed pixel ball game logic in Verilog for Basys FPGA; implemented collision detection and physics simulation; achieved 60 FPS
- Engineered frequency detection system in Verilog for musical note recognition on Icebreaker FPGA; implemented FFT-based pipeline, achieving 99.2% accuracy

TECHNICAL SKILLS

Languages: C, C++, Python, SystemVerilog, Verilog, VHDL, RISC-V Assembly, Java

Developer Tools: Vivado, Yosys, Quartus, Synopsys VCS, Cadence Virtuoso, Git, STM32CubeIDE, PlatformIO, Arduino IDE, Flutter

Technologies/Frameworks: ASIC/FPGA: RTL, Synthesis, Static Timing Analysis (STA); Verification: Constrained-Random, UVM (basics); RISC-V

Digital Design: RTL Design, Testbench Development, Hardware Verification Methodologies, Timing Closure, Design Constraints (SDC), Verification Planning

Embedded/Microcontrollers: ESP32, STM32, AVR, FreeRTOS, ESP-IDF, I2C, SPI, UART, board bring-up, sensor integration